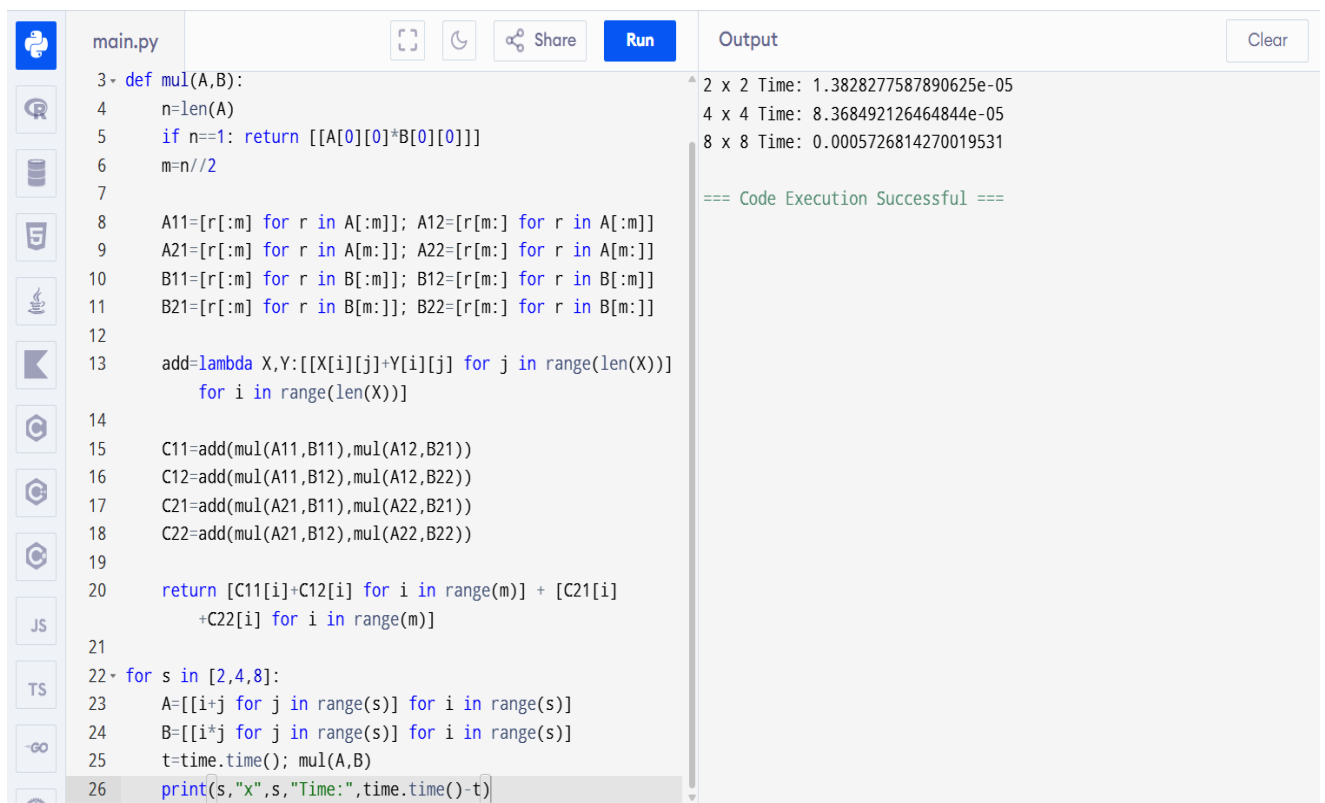


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EXPERIMENTS

Evaluate divide and conquer matrix multiplication by implementing recursive splitting of matrices. Record execution time for different matrix sizes. Analyze how problem division reduces complexity. Interpret results and justify the efficiency of recursive approaches.



The screenshot displays a Jupyter Notebook with a file named 'main.py'. The code implements a recursive matrix multiplication algorithm. It defines a function 'mul(A,B)' that splits a matrix into four quadrants, recursively multiplies them, and then combines the results. The base case is when the matrix size is 1. The code also includes a lambda function 'add' for adding two matrices. The main execution part iterates over matrix sizes [2, 4, 8], creates matrices A and B, measures the execution time, and prints the results.

```
3- def mul(A,B):
4     n=len(A)
5     if n==1: return [[A[0][0]*B[0][0]]]
6     m=n//2
7
8     A11=[r[:m] for r in A[:m]]; A12=[r[m:] for r in A[:m]]
9     A21=[r[:m] for r in A[m:]]; A22=[r[m:] for r in A[m:]]
10    B11=[r[:m] for r in B[:m]]; B12=[r[m:] for r in B[:m]]
11    B21=[r[:m] for r in B[m:]]; B22=[r[m:] for r in B[m:]]
12
13    add=lambda X,Y:[[X[i][j]+Y[i][j] for j in range(len(X))]
14                    for i in range(len(X))]
15
16    C11=add(mul(A11,B11),mul(A12,B21))
17    C12=add(mul(A11,B12),mul(A12,B22))
18    C21=add(mul(A21,B11),mul(A22,B21))
19    C22=add(mul(A21,B12),mul(A22,B22))
20
21    return [C11[i]+C12[i] for i in range(m)] + [C21[i]
22          +C22[i] for i in range(m)]
23
24 for s in [2,4,8]:
25     A=[[i+j for j in range(s)] for i in range(s)]
26     B=[[i*j for j in range(s)] for i in range(s)]
27     t=time.time(); mul(A,B)
28     print(s,"x",s,"Time:",time.time()-t)
```

The Output section shows the execution results for matrix sizes 2x2, 4x4, and 8x8, along with a success message:

```
2 x 2 Time: 1.3828277587890625e-05
4 x 4 Time: 8.368492126464844e-05
8 x 8 Time: 0.0005726814270019531
=== Code Execution Successful ===
```